Multistable active phases of complex contagions from temporal higher-order interactions

Santiago Lamata-Otín,[1,2,3] Federico Malizia,[4] Leah A. Keating,[3,5] Guillaume St-Onge,[6,7] Vito Latora,[8,9,10] Jesús Gómez-Gardeñes,[1,2] and Laurent Hébert-Dufresne[3,5,11,12]

[1]*Department of Condensed Matter Physics, University of Zaragoza, 50009 Zaragoza, Spain*
[2]*GOTHAM lab, Institute of Biocomputation and Physics of Complex Systems (BIFI), University of Zaragoza, 50018 Zaragoza, Spain*
[3] *Vermont Complex Systems Institute, University of Vermont, Burlington, VT 05405 USA*
[4]*Department of Network and Data Science, Central European University, Vienna, Austria*
[5]*Department of Computer Science, University of Vermont, Burlington, VT 05405 USA*
[6]*Laboratory for the Modeling of Biological and Socio-technical Systems, Northeastern University, Boston MA, USA*
[7]*The Roux Institute, Northeastern University, Portland ME, USA*
[8]*Department of Physics and Astronomy, University of Catania, 95125 Catania, Italy*
[9]*School of Mathematical Sciences, Queen Mary University of London, London E1 4NS, United Kingdom*
[10]*Complexity Science Hub Vienna, A-1080 Vienna, Austria*
[11]*Complexity Science Hub, Vienna, Austria*
[12]*Santa Fe Institute, Santa Fe, NM 87501, USA*

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Higher-order contagion models show that group-based reinforcement can cause a discontinuous transition from an inactive to an active phase, yet these results are typically derived on static hypergraphs or with mean-field approximations that neglect temporal dynamics. Here, we demonstrate that temporality is essential using an analytically tractable approximate master equation model that incorporates turnover in group composition. Our theory interpolates between quenched and mean-field limits and reveals that increasing temporality can induce discontinuous transitions, enlarge the bistable region, and generate a nonmonotonic invasion threshold, implying that excessive temporality can suppress contagion. When real-world temporal turnover rates and group-size heterogeneity are incorporated, four distinct dynamical phases emerge, combining continuous or discontinuous transitions with localized or delocalized activity. In localized regimes, we uncover multistable active phases with multiple coexisting active states, which are observed in neither the annealed nor the quenched limits, and extend classical absorbing-active bistability. Finally, we demonstrate that the emergence of discontinuous transitions in real-world settings requires stronger nonlinear reinforcement than previously thought, indicating that simulations in static structures can yield qualitatively misleading predictions.

In social systems, the adoption of new behaviors or ideas can require more than a single exposure: individuals may need reinforcement from multiple peers before changing state [1–3]. This feature has motivated a broad class of complex contagion models [4–12] traditionally based on threshold rules [4–9] or on nonlinear synergistic mechanisms [10–12]. Recent work by Iacopini et al. [13] revitalized this field by representing reinforcement through a simple interaction structure: groups. Even when operating only over groups, their higher-order contagion formulation showed that group reinforcement can change the nature of the transition from a no-adopter to an active adoption state, from continuous to discontinuous as per previous models [6, 12]. This result spurred developments across diverse dynamical processes specified over groups, including synchronization [14–18], self-organization [19], and game theory [20–22].

Real group interactions are inherently temporal [23]: individuals can join and leave groups on time scales slower than, comparable to, or faster than contagion itself. Mathematical descriptions of discontinuous transitions typically adopt the annealed (mean-field) limit [13, 24], corresponding to infinitely fast reshuffling. Under a homogeneous mean-field approximation, the invasion threshold, i.e., the critical condition under which a small seed of adopters can successfully spread, is independent of the nonlinearity of the contagion dynamics [13]. However, when the temporal evolution of the structure is explicitly incorporated, this picture may change. For linear contagion, temporality is known to shift the invasion threshold towards lower values by enhancing mixing (except if driven by hub localization in heterogeneous structures) [25, 26], while in the opposite limit of a static (quenched) structure, the threshold can even diverge as the network becomes fragmented [27].

The limited efforts aimed at understanding higher-order contagion over temporal groups report a conflicting trend. Using a Markovian model, Chowdhary et al. [28] showed that the threshold shifts towards larger values on time-varying group-based structures, even when contagion is linear. This contradicts previous results [25, 26], raising the question of whether group-based contagions are fundamentally different or whether this discrepancy arises from how temporality is incorporated. Beyond how temporality shifts critical thresholds, it remains unclear how it affects the active phase in general.

In this work, we develop an analytically tractable approximate master equations (AMEs) model that couples higher-order contagion with temporal variability, encoded through a plasticity rate. This framework leads us to three important results. (*i*) We find that increasing plasticity can qualitatively change the nature of the phase transition, transforming a continuous transition into a discontinuous one and enlarging the bistable region in systems with homogeneous group-size distributions. In this setting, we derive analytically that the invasion threshold becomes non-monotonic in the plasticity rate, implying the existence of an optimal plasticity rate beyond which temporality suppresses adoption. (*ii*) We find that introducing the group-size heterogeneity characteristic of real-world systems reorganizes the phase portrait into four distinct dynamical regimes, distinguished by the interplay between nonlinear reinforcement and mesoscopic localization. In particular, we observe a region displaying a hybrid continuous transition with active bistability, as well as a region with a discontinuous hybrid transition exhibiting three coexisting stable states: the absorbing state and two active states. (*iii*) We observe that the empirically measured temporal and structural features of empirical social systems require extremely strong nonlinear reinforcement to exhibit discontinuous transitions. This result demonstrates that static or mean-field representations may yield qualitatively misleading predictions.

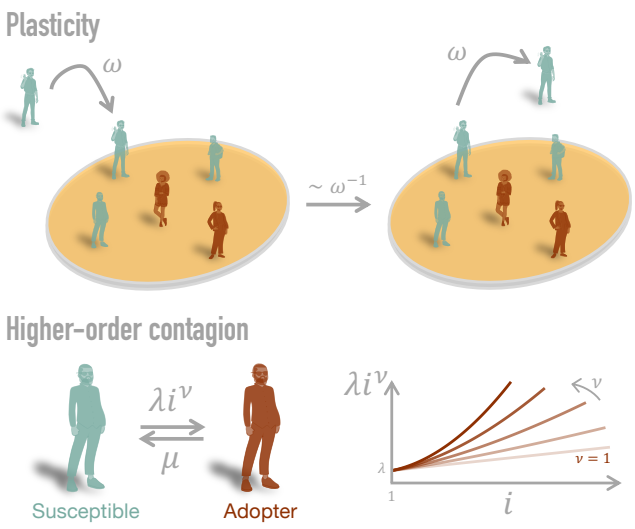

FIG. 1: **Temporal higher-order contagion model.** Individuals reshuffle between groups at rate ω (plasticity), while contagion unfolds within groups: a susceptible in a group of size n with i adopters becomes infected at rate $\beta(n,i) = \lambda i^{\nu}$, and adopters recover at rate μ. The synergy exponent ν controls nonlinear reinforcement.

Model of temporal higher-order contagion

To capture the impact of temporality on higher-order contagion, we construct a model in which all changes in group composition are coarse-grained into a single node-level group switching process occurring at rate ω. We consider (*i*) a set of groups whose size n follows the distribution $\{p_n\}$, and (*ii*) a population of individuals whose number of group memberships k follows the distribution $\{g_k\}$. Two processes govern the dynamics (see Fig. 1). *Higher-order contagion*: a susceptible individual in a group of size n with i adopters becomes an adopter at rate $\beta(n,i) = \lambda i^{\nu}$, where λ is the intrinsic adoption rate and ν controls the strength of synergistic reinforcement; adopters recover at rate μ [29, 30]. *Plasticity*: random individuals are swapped between random groups at rate ω.

The resulting coevolutionary dynamics are described by a set of AMEs, Eqs. (8)-(11) in Methods. The model tracks the density of susceptible individuals with membership k, s_k, and the fraction of groups of size n containing i adopters, $f_{n,i}$. From these quantities, we compute the stationary fraction of adopter population $I^{\star}$ (see Eq. (12) in Methods), which we use throughout the following sections to characterize the macroscopic behavior of temporal higher-order contagion. We refer to Eqs. (13)-(14) in Methods and Supplementary Note for details on how the model is solved.

Temporality reshapes the phase transition of higher-order contagion

Temporality fundamentally reorganizes the phase transition of higher-order contagion. We study in Fig. 2 the case where individuals can effectively participate in only one face-to-face interaction at a time ($g_k = \delta_{k,1}$). In the quenched limit, $\omega = 0$, group composition is frozen, and individuals remain confined to their initial interaction patterns. Under these conditions, contagion cannot invade: adopters are not able to sustain activity due to the finite size of groups, and the system always collapses into the absorbing state $I^{\star} = 0$ [27]. The invasion threshold λ_c, i.e., the critical condition under which a small seed of adopters can successfully spread, therefore diverges (see Supplementary Notes 1, 2 and 3 for the derivation),

$$\lim_{\omega \to 0} \lambda_c(\omega) = \infty \tag{1}$$

revealing that no transition exists in fragmented structures.

Any degree of temporal evolution ($\omega > 0$) changes this picture qualitatively. Even low plasticity rates expose individuals to multiple group configurations over time. As a

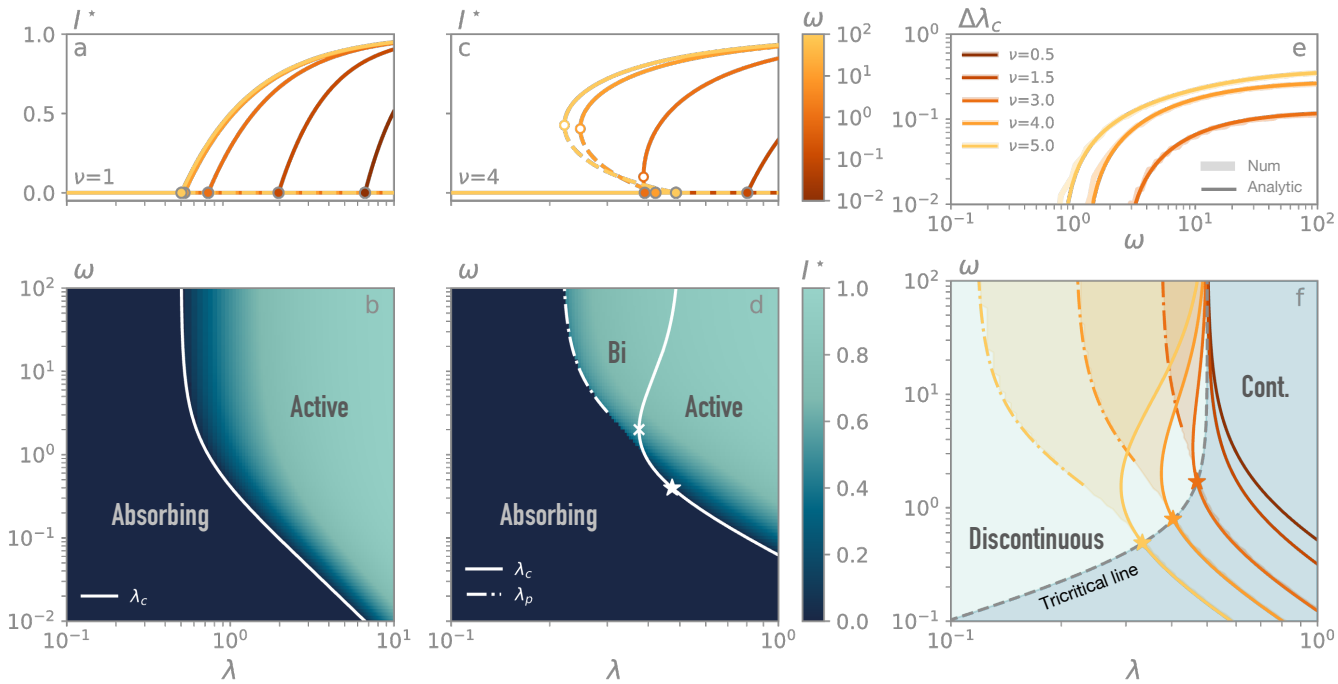

FIG. 2: **Temporality reshapes the critical behavior of higher-order contagion. a–b** Stationary prevalence $I^\star$ and corresponding phase diagram for linear contagion ($\nu = 1$) as a function of the spreading rate λ and the plasticity rate ω. Increasing temporal turnover lowers the invasion threshold, and the transition remains continuous. **c–d** Same representation for nonlinear contagion ($\nu = 4$). Here, temporality fundamentally alters the transition: the invasion threshold becomes non-monotonic in ω, and sufficiently strong synergy produces a discontinuous transition. Analytical predictions for the invasion threshold $\lambda_c(\omega)$ and the persistence threshold $\lambda_p(\omega)$ are shown throughout, the tricritical point derived using Eq. (16) in Methods is marked with a star in panel d, and the minimum of the invasion threshold according to Eq. (4) is marked with a cross in panel d. **e** Width of the bistable region, $\Delta\lambda$, as a function of the plasticity rate for different values of the synergy exponent ν. **f** Bifurcation diagram showing the analytically derived tricritical line separating continuous and discontinuous transitions, together with the corresponding invasion and persistence thresholds for the nonlinearities considered in panel e. All analytical curves closely match the steady-state solutions of the AMEs. In all panels, each individual belongs only to one group ($g(k) = \delta_{k,1}$), the groups have homogeneous size ($p(n) = \delta_{n,1}$), and $\mu = 1$.

result, a transition from the absorbing to the active phase emerges. In the opposite limit of infinitely fast reshuffling ($\omega \to \infty$), individuals effectively sample all possible group realizations, yielding the mean-field critical point

$$\lim_{\omega\to\infty} \lambda_c(\omega) = \frac{\mu \langle n \rangle}{\langle k \rangle \langle n(n-1) \rangle}, \tag{2}$$

which holds for both linear and nonlinear contagion (see Supplementary Note 3 for the derivation). The full curve interpolating between these limits is derived in the Supplementary Note 2 and detailed in Eq. (15) in Methods, and matches the steady-state AME solutions in Fig. 2a–d.

Having established that temporality creates a transition that is absent in the quenched limit, we next examine how it changes the nature of this transition. When contagion is linear ($\nu = 1$), the transition is always continuous: $I^\star$ grows smoothly from zero as λ crosses λ_c. However, in the presence of nonlinear reinforcement, i.e. $\nu > 1$, temporality can fundamentally alter the order of the phase transition. As shown in Fig. 2c-d, for sufficiently strong synergy, increasing the plasticity rate causes the continuous transition to change to a discontinuous one accompanied by bistability, in which both the absorbing and active states are locally stable. The tricritical condition producing the star marker in Fig. 2d, which separates the regime of continuous transition with the regime of discontinuous transition and bistability, is derived in the Supplementary Note 2 and is reported in Eq. (16) in Methods. As shown in Fig. 2e-f, the width of the bistable region increases with both the plasticity rate ω and the synergy exponent ν, revealing that temporal turnover enhances the reinforcement effects needed to generate explosive adoption events [31].

Once this change in the phase transition order is understood, a second consequence of temporality becomes clear: in the nonlinear regime $\nu > 1$, the invasion threshold $\lambda_c(\omega)$ becomes non-monotonic in the plasticity rate. Fig. 2c–d show that $\lambda_c(\omega)$ initially decreases as temporal mixing increases, reaching a finite minimum

at $\omega^\star$, after which it increases again. Thus, temporality enhances contagion only up to an optimal mixing rate; beyond this point, additional reshuffling suppresses the reinforcement required for adoption.

This non-monotonicity in homogeneous structures is characteristic of higher-order interactions. When only pairwise interactions are present ($p_n = \delta_{n,2}$), the critical line reduces to $\lambda_{c,n=2}(\omega) = \mu + \mu^2/\omega$, which is strictly monotonic and independent of ν. In contrast, for structure consisting only of triplets ($p_n = \delta_{n,3}$), the critical line reads

$$\lambda_{c,n=3}(\omega) = \frac{\mu+\omega}{2^\nu}\left[\sqrt{1+\frac{2^\nu\mu}{\omega}}-1\right], \tag{3}$$

from which a finite minimum emerges only when $\nu > 2$. The corresponding optimal plasticity rate and minimal threshold are

$$\omega^\star_{n=3} = \mu\,\frac{2^{\nu/2}}{2^{\nu/2}-2}, \qquad \lambda^\star_{c,n=3} = \mu\,\frac{2^{\nu/2}-1}{2^{\nu-1}}. \tag{4}$$

In Eqs. (17)-(19) and Supplementary Note 3, we present the general form of Eqs. (3)-(4) beyond $g_k = \delta_{k,1}$, together with their derivation. Allowing agents to participate in multiple groups in parallel ($g_k = \delta_{k,\kappa},\ \kappa > 1$) defines the transition even in the absence of temporality and shifts the minimum of the invasion threshold toward zero, potentially pushing it outside the physical domain. As a result, $\lambda_c(\omega)$ becomes strictly increasing in ω. In this regime, increasing plasticity consistently suppresses adoption, reversing the facilitation observed in linear dynamics. Larger typical group sizes have a similar effect in the invasion threshold, as they amplify synergistic reinforcement (see Supplementary Fig. 2 in Supplementary Note 4 for further details).

In contrast, the persistence threshold, i.e., the critical condition below which an established active state cannot be sustained (and therefore, the left boundary of the bistability region), remains strictly monotonic in ω. This occurs because, when starting from a fully adopter population, groups contain a sufficient density of adopters such that plasticity cannot disrupt the reinforcement needed to sustain adoption. Instead, reshuffling primarily increases contacts between susceptible and adopter individuals. For large plasticity rates and $p_n = \delta_{n,3}$, we derive in the Supplementary Note 5 the closed expression reported in Eqs. (20)-(23) in Methods; its asymptotic value is

$$\lim_{\omega\to\infty} \lambda_{p,n=3}(\omega) = \mu\frac{4\left(2^\nu-2\right)}{2^{2\nu}}, \tag{5}$$

and therefore, accounting for Eq. (2), the width of the bistable region approaches

$$\lim_{\omega\to\infty} \Delta\lambda_{n=3}(\omega) = \frac{\mu}{2}\left(1-2^{2-\nu}\right)^2. \tag{6}$$

As shown in Fig. 2, these analytical expressions match the stationary solutions of the AMEs.

New phase transitions and multistability from temporality and heterogeneity

Having established how plasticity reshapes higher-order contagion in homogeneous synthetic settings, we now turn to real-world social systems from various contexts [32–38], where group sizes are heterogeneous. Each dataset is characterized by a group-size distribution $\{p_n\}$ and a membership distribution $\{g_k\}$ (see Supplementary Fig. 2 in Supplementary Note 6), the latter being $g_k \approx \delta_{1,k}$ across all contexts.

These structural parameters determine the tricritical line $[\lambda^\star_c, \nu^\star_c](\omega)$ separating continuous and discontinuous transitions. For homogeneous group sizes, the tricritical line is a monotonic function of the plasticity rate (Fig. 2f). However, once we incorporate the empirical size and membership distributions of a real system, the tricritical boundary becomes non-monotonic and develops a folded structure (see Fig. 3a for the high-school dataset [32], and Supplementary Fig. 3a in Supplementary Note 7 for the other datasets). As a consequence, for a fixed plasticity rate, the tricritical condition in Eq. (16) can admit multiple solutions: for instance, at $\omega = 5$ the horizontal cut intersects the tricritical curve three times.

To understand the dynamical consequences, in Fig. 3b we show the stationary prevalence as a function of the intrinsic adoption rate for seven values of the synergy exponent ν, with the corresponding invasion thresholds indicated in Fig. 3a. Continuous segments of the curves correspond to linearly stable stationary states, whereas dashed segments denote unstable solutions. As nonlinearity increases, the system undergoes four qualitatively distinct dynamical regimes, illustrated in Fig. 3c–f.

For weak reinforcement $\nu \approx 1$, the transition is continuous and resembles classic SIS dynamics (Fig. 3c). At intermediate ν, reinforcement within groups becomes strong enough to induce a discontinuous transition with bistability between absorbing and active states (Fig. 3d). As the synergy exponent ν increases further, the qualitative structure of the phase diagram changes again. In Fig. 3e–f, the active phase no longer consists of a single endemic state. Instead, multiple active branches appear. In Fig. 3e we observe a hybrid continuous transition with active bistability: activity emerges continuously from zero, yet two stable active branches coexist over a finite parameter range. For even stronger synergy (Fig. 3f), the transition becomes a discontinuous hybrid one, with three coexisting stable states, namely the absorbing state and two distinct active states.

To understand the emergence of these active states, we introduce the effective participation ratio $P =$

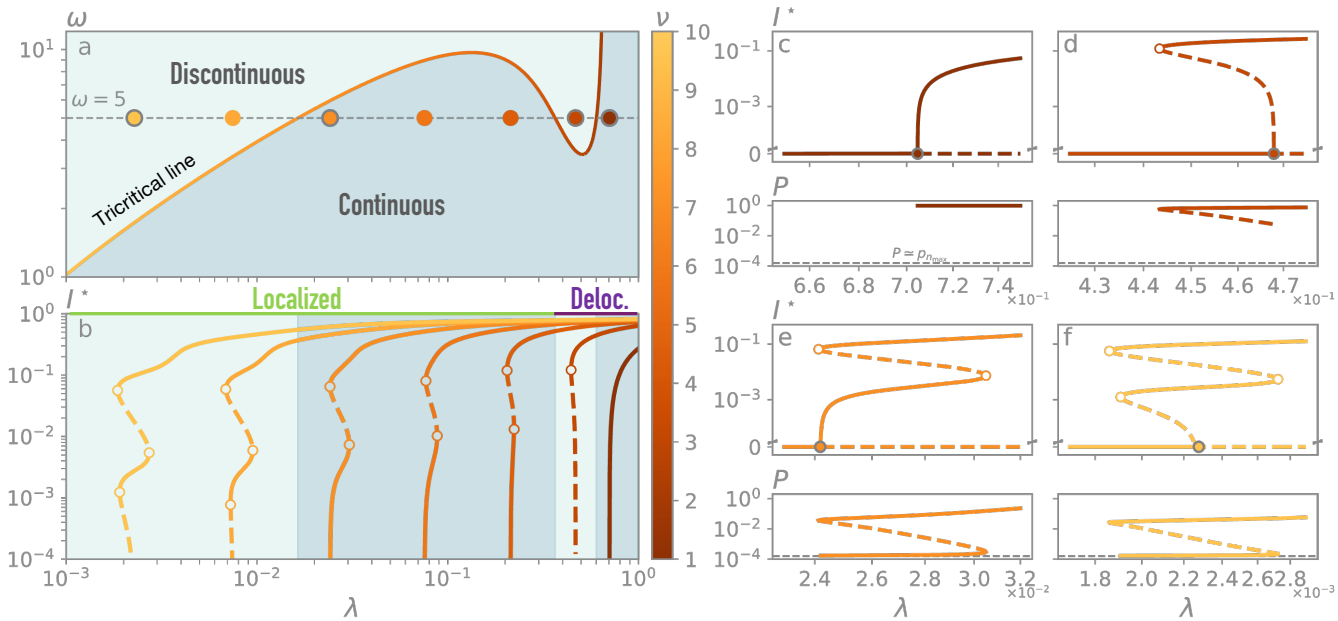

FIG. 3: **Interplay between temporality and heterogeneity yields multistability.** **a** Tricritical line in the (ω, λ) plane for the empirical group-size and membership distributions of a high school [32]. **b** Stationary prevalence $I^\star$ as a function of the adoption rate λ for increasing values of the synergy exponent ν (color-coded). Solid (dashed) lines denote stable (unstable) stationary states. **c-f** Representative phase diagrams illustrating four distinct dynamical regimes: continuous transition in panel **c**, discontinuous transition with absorbing–active bistability in panel **d**, continuous hybrid transition with active bistability in panel **e**, and discontinuous hybrid transition with three coexisting stable states in panel **f**. Bottom rows show the effective participation ratio P quantifying the degree of localization of activity across groups.

$\left(\sum_n p_n I_n^2\right)^2 / \sum_n p_n I_n^4 \in [p_{n_{\max}}, 1]$ to quantify the effective number of groups contributing to adoption: it is large when activity is broadly distributed across groups and small when it is concentrated in a limited subset (see bottom rows in Fig. 3c–f) [30]. While the active states in Fig. 3c–d correspond to delocalized activity, the ones in Fig. 3e–f exhibit reduced participation ratios, signaling mesoscopic localization. In particular, the lower stable state in both panels is sustained by the adopters in the largest groups, whereas the upper state remains only partially localized.

In Supplementary Fig. 3b we further assess the role of group-size heterogeneity using synthetic structures. Supplementary Fig. 4 complements this analysis by presenting the phase diagram analogous to Fig. 3b in the quenched and annealed limits. The quenched regime displays continuous transitions with non-monotonic prevalence growth and mesoscale localization plateaus [30, 39, 40], whereas the annealed regime exhibits the discontinuous transitions predicted by mean-field analysis [13]. Importantly, multistable active phases with multiple coexisting active states are observed in neither limit, demonstrating that multistability is an intrinsically temporal phenomenon that emerges only for finite mixing rates.

Temporal reshuffling determines the transition class in empirical systems

As shown in the previous section, the structural distributions $\{p_n\}$ and $\{g_k\}$ define a tricritical line separating regions with continuous and discontinuous transitions as the plasticity rate varies (see Fig. 3a). In Fig. 4 we represent this tricritical line in the $\nu\lambda$-space for each real-world structure, illustrating how the tricritical point moves continuously from the quenched (static) to the annealed (mean-field) limit as ω increases. Across all empirical settings, discontinuous transitions occur only for consistently large nonlinearities $\nu_c^\star$, highlighting that strong synergistic reinforcement is necessary for bistability.

However, locating the actual empirical systems along the tricritical line requires incorporating their intrinsic temporal scales. The transition class is not determined by structural heterogeneity alone, but by the competition between the contagion timescale (μ^{-1}) and the plasticity timescale (ω^{-1}). To quantify the latter, we introduce the effective plasticity rate $\langle\omega\rangle$, defined as the inverse of the average residence time of individuals within groups (see Eq. (7) in Methods and Supplementary Note 6).

We therefore report in Fig. 4 the tricritical point corresponding to the empirical value $\langle\omega\rangle$ for several contagion timescales (with low (large) μ values corresponding to slow (fast) dynamics). When contagion is fast com-

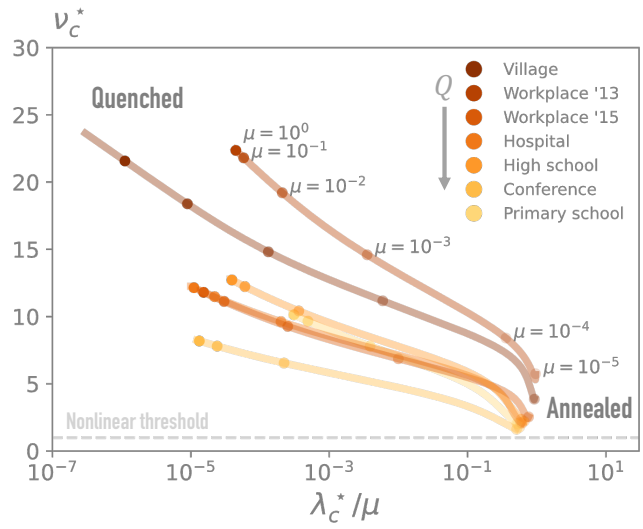

FIG. 4: **Tricritical point across dynamical timescales in real-world systems.** For each dataset [32–38], colored curves trace how the critical point shifts as the plasticity rate ω is varied from the quenched to the annealed limit; markers indicate the tricritical points corresponding to the empirically measured plasticity rate $\langle\omega\rangle$ for six contagion timescales, from $\mu = 10^0$ to $\mu = 10^{-5}$. Colors indicate the rank of the effective structural coupling Q introduced in Eq. (24).

pared to reshuffling ($\mu \gg \omega$), the structure is effectively quenched: reinforcement persists within stable groups of adopters, and discontinuous transitions require large nonlinearities $\nu_c^\star$. Conversely, when structure evolution dominates ($\omega \gg \mu$), the system approaches the annealed regime where individuals experience many group configurations during the adoption period, and $\nu_c^\star$ is reduced.

Finally, to understand the differences between real-world systems, we rank them according to their effective structural coupling Q (see Eq. (24) in Methods), a descriptor of the effective inter-group connectivity experienced during contagion. By construction, Q makes explicit that plasticity enhances inter-group connectivity by increasing effective mixing across groups over the contagion timescale. As shown in Fig. 4, weakly coupled systems require stronger nonlinearities to undergo discontinuous transitions, whereas strongly coupled systems transition at lower reinforcement levels.

Together, these results show that the macroscopic transition class is governed by the competition of timescales between contagion and structure evolution. Therefore, temporality is not a secondary correction to structure, but a primary control parameter, and static aggregated or annealed mean field representations can qualitatively mischaracterize both the nature and location of the critical boundary.

Discussion

Temporality plays a fundamental role in shaping collective dynamics across a wide range of complex systems. While its impact has been explored in pairwise interaction networks, its consequences for group-based structures remain less understood. Here, we introduced an analytically tractable model accounting for temporality through a plasticity rate that captures how individuals change groups.

We uncovered how temporality reorganizes the entire phase diagram of higher-order contagion dynamics: it determines whether the phase transition from the absorbing to the active state is continuous, discontinuous, or altogether absent, and induces a non-monotonic invasion threshold characterized by an optimal plasticity rate. This behavior mirrors the detriment effect of mobility observed in metapopulation epidemics [41, 42]. Introducing the group-size heterogeneity characteristic of real-world systems further enriches this picture. The interplay between temporality and heterogeneity yields multiple active states that differ in their degree of mesoscopic localization across groups. Therefore, new phases emerge, displaying hybrid transitions with continuous onset but active bistability, or discontinuous transitions with three coexisting stable states, both beyond classical absorbing–active bistability. This entire behavior does not occur in either static structures or mean-field approximations, demonstrating that temporality is not a correction to the structure but a primary dynamical ingredient. Although multistability has previously been reported in critical-mass dynamics, it was driven by the community structure of the static hypergraphs [43].

Our results on real-world structures reveal that discontinuous transitions require substantially stronger nonlinear reinforcement than typically obtained from aggregated and static representations. Such aggregation suppresses temporal features [44–46] and can artificially alter hyperedge overlap and density [18, 47–50]. As a consequence, both the location and qualitative nature of the transition may be mischaracterized when temporality is neglected.

Another outcome of this work is the reconciliation of seemingly contradictory results. For linear contagion in homogeneous structures, temporality shifts the invasion threshold towards lower values by enhancing mixing [25, 26]. This trend can reverse in higher-order contagion, but only under sufficiently strong nonlinear reinforcement or large group sizes. In these scenarios, rapid reshuffling prevents individuals from remaining long enough within stable groups to accumulate reinforcement, thereby hindering adoption. Supplementary Fig. 5 in Supplementary Note 8 shows that the consistent suppression of contagion reported by Chowdhary

et al. [28] arises when the reshuffling process does not preserve the degree sequence.

Several avenues for future work emerge. Although the role of temporality has been explored in consensus dynamics [51] and evolutionary game theory [22, 52], its role in shaping phase transitions across other dynamical processes remains largely unexplored. Moreover, plasticity itself may also be adaptive: individuals modify their group participation in response to contagion [19, 53–55] or in response to group traits [56, 57], feedback that could be incorporated into solvable extensions of the present AMEs framework. More broadly, combining higher-order contagion with temporal heterogeneity in transmission, memory, or behavioural responses may yield richer classes of dynamical transitions. Beyond these theoretical directions, our results raise methodological implications: aggregated hypergraphs can significantly misrepresent dynamical behaviour, and future empirical studies should integrate temporal information whenever possible.

Together, these results establish a unified framework for temporal higher-order contagion, and reveal that temporality is not a secondary correction to higher-order contagion, but a primary mechanism that determines the macroscopic transition class. Ignoring temporality fundamentally mischaracterizes the nature and location of the critical transition, highlighting the need to treat time as an intrinsic part of the interaction structure.

METHODS

Definition of the effective plasticity rate

To quantify the reshuffling process, we measure the average residence time of individuals within groups. Let τ_e denote the time interval between two consecutive group-change events involving the same individual (see Fig. 1a), and let E be the total number of such events in the dataset. We define the effective plasticity rate as the frequency of individual movement, computed as the inverse of the average inter-event time:

$$\langle \omega \rangle = \frac{1}{\langle \tau \rangle}, \qquad \langle \tau \rangle = \frac{1}{E}\sum_{e=1}^{E} \tau_e, \tag{7}$$

where an event is defined as the instantaneous jump of an individual from one group to another. This quantity provides a direct measurement of how rapidly group composition changes in real systems. We refer the reader to the Supplementary Note 6 for further details.

wAMEs model of complex contagion with plasticity

We follow the framework of St-Onge et al. [30], a non-linear generalization of the group-based Approximate Master Equations (AMEs) [27], where we track the evolution of the susceptible population with membership k, denoted s_k, and the fraction of groups of size n with i infectious individuals within, denoted $f_{n,i}$. Adding the mechanism for group plasticity at rate ω, the governing equations read:

$$\frac{ds_k}{dt} = \mu(1-s_k) - krs_k \tag{8}$$

$$\begin{aligned}\frac{df_{n,i}}{dt} &= (i+1)\big(\mu+\omega(1-I)\big)\, f_{n,i+1} \\ &\quad - \left[i(\mu+\omega(1-I)) + (n-i)(\beta(n,i)+\rho+\omega I)\right] f_{n,i} \\ &\quad + (n-i+1)(\beta(n,i-1)+\rho+\omega I) f_{n,i-1}.\end{aligned} \tag{9}$$

Eq. (8) describes the change in the fraction of susceptible individuals with k group memberships. This fraction increases when infected individuals recover and decreases when susceptibles become infected through exposure in any of the k groups to which they belong.

The terms in Eq. (9) capture all transitions that modify the number of infectious individuals within a group of size n. The first term represents transitions from state $i+1$ to i, resulting either from the recovery of an infectious individual at rate μ or from its being swapped with a susceptible one at rate ω. The second term accounts for all events that remove probability mass from $f_{n,i}$: any of the i infectious individuals may recover at rate μ or be swapped out at rate ω, and any of the $n-i$ susceptible individuals may become infected through within-group transmission at rate $\beta(n,i)$, external influence at rate ρ, or by being swapped with an infectious individual at rate ω. The third term describes transitions from state $i-1$ to i, occurring when one of the $n-(i-1)$ susceptibles becomes infectious via internal transmission at rate $\beta(n,i-1)$, external influence at rate ρ, or by being swapped with an infectious individual at rate ω. Importantly, the external influence ρ represents the mean-field infection pressure exerted by all groups to which a susceptible node belongs, excluding the focal group, and is given by

$$\rho(r) = r \cdot \frac{\sum_k k(k-1)s_k g_k}{\sum_k k s_k g_k}, \tag{10}$$

which corresponds to the product of the mean-field infection rate coming from a randomly selected external group and the mean excess membership of a susceptible node. In other words, if we pick a susceptible node in a given group, the prefactor represents the expected number of other groups to which it belongs. The mean-field infection rate itself is defined as

$$r = \frac{\sum_{n,i} \beta(n,i)(n-i) f_{n,i} p_n}{\sum_{n,i}(n-i) f_{n,i} p_n}, \tag{11}$$

which is the average value of $\beta(n,i)$ times the susceptible members within a group over the distribution of infected nodes. In addition, the global prevalence,

$$I(t) = \sum_k (1 - s_k(t) g_k), \tag{12}$$

serves as the main order parameter of the system. Together, these contributions provide a compact characterization of the gain and loss processes that govern the evolution of $f_{n,i}$.

Stationary states

In the stationary state, the whole system can be rewritten entirely in terms of the variables r, ρ and I (See Supplementary Note 1). We therefore define the functional

$$\mathcal{M}\left[\rho(r), I(r)\right] = \frac{\sum_{n,i} \beta(n,i)(n-i) f_{n,i}(\rho, I) p_n}{\sum_{n,i} (n-i) f_{n,i}(\rho, I) p_n}, \tag{13}$$

where the dependency on ρ and I enters through the stationary group-state distributions $f_{n,i}(\rho, I)$. At equilibrium, the system must satisfy the implicit self-consistency condition

$$r = \mathcal{M}\left[\rho(r), I(r)\right]. \tag{14}$$

This relation can be solved numerically to obtain the fixed points of the dynamics. To determine their stability, we additionally evaluate the Jacobian of the stationary system, as detailed in the Supplementary Note 1.

Invasion threshold

The invasion threshold corresponds to the critical points that mark the limit of the domain of validity of the solution $r = \mathcal{M}\left[\rho(r), I(r)\right]$, arising when r is tangent to $\mathcal{M}\left[\rho(r), I(r)\right]$, i.e. when $\left.\frac{\partial M}{\partial r}\right|_{r\to 0} = 1$. As derived in the Supplementary Note 2, the condition reads

$$\begin{aligned} 1 \;=\; & \left\langle \sum_{i=1}^{n} \frac{n!}{(n-i-1)!\, i!} \left(\frac{\lambda_c}{\mu+\omega}\right)^i \prod_{j=1}^{i} j^\nu \right\rangle \\ & \times \left(\frac{\langle k(k-1)\rangle}{\langle k\rangle\langle n\rangle} + \frac{\omega}{\mu}\frac{\langle k\rangle}{\langle n\rangle} \right). \end{aligned} \tag{15}$$

Bistability threshold

To fully characterize the phase portrait, we derive in the Supplementary Note 2 the condition for the tricritical point that separates the absorbing, active, and bistable regions. Imposing that $\left.\frac{\partial^2 M}{\partial r^2}\right|_{r\to 0} = 0$ we reach that the condition reads

$$\begin{aligned} 0 \;=\; & F \frac{\langle k(k-1)\rangle^2}{\langle k\rangle^2} + 2G \frac{\langle k(k-1)\rangle}{\langle k\rangle}\frac{\langle k\rangle}{\mu} + H\frac{\langle k\rangle^2}{\mu^2} \\ & + 2\frac{\frac{1}{\mu}\Big(\langle k^2\rangle^2 - \langle k^3\rangle\langle k\rangle\Big) - \frac{\omega}{\mu^2}\langle k^2\rangle\langle k\rangle^2}{\langle k(k-1)\rangle\langle k\rangle + \frac{\omega}{\mu}\langle k\rangle^3}. \end{aligned} \tag{16}$$

Here, $F \equiv F\left[p_n, \beta(n,i), \omega\right]$, $G \equiv G\left[p_n, \beta(n,i), \omega\right]$, $H \equiv H\left[p_n, \beta(n,i), \omega\right]$ are specified in the Supplmentary Note 2.

Optimal plasticity rate to maximize adoption

The general expression for the critical line given $g(\kappa) = \delta_{\kappa,k}$ is derived in the Supplementary Note 3 and reads

$$\lambda_c(\omega; k) = \frac{\mu+\omega}{2^\nu}\left[\sqrt{1 + \frac{2^\nu \mu}{k\omega + \mu(k-1)}} - 1\right], \tag{17}$$

which again reduces to Eq. (15) for $k = 1$. For $\nu > 2$, differentiating Eq. (17) yields the position of the minimum:

$$\omega^\star(k) = \mu\, \frac{2^{\nu/2+1} + 2^{\nu+1} - 4 - k(2^\nu - 4)}{k(2^\nu - 4)}, \qquad (\nu > 2), \tag{18}$$

which reduces to Eq. (4) for $k = 1$. As k increases, the minimum $\omega^\star(k)$ shifts toward zero and eventually becomes non-positive. When this happens, the minimum disappears entirely and $\lambda_c(\omega; k)$ becomes strictly increasing in ω.

Evaluating Eq. (17) at $\omega^\star(k)$ gives the minimal threshold

$$\lambda_c^\star(k) = \frac{1}{k}\,\lambda_c^\star(1) = \frac{\mu}{k}\,\frac{2^{\nu/2} - 1}{2^{\nu-1}}. \tag{19}$$

Persistence threshold

The expression of the persistence threshold $\lambda_p(\omega)$ for large values of ω, $g(k) = \delta_{k,1}$ and $p(n) = \delta_{n,3}$ is obtained in the Supplementary Note 5, and reads

$$\lambda_p(\omega) = \frac{-B_p(\omega) + \sqrt{B_p(\omega)^2 - 4A_p(\omega)C_p(\omega)}}{2A_p(\omega)}, \tag{20}$$

where

$$A_{\rm p}(\omega, \mu) = 2^{\nu+1}\big(2^\nu \omega + 3\mu\big), \tag{21}$$

$$B_{\rm p}(\omega, \mu) = 4^\nu \omega^2 + 2^{\nu+1}\mu\,\omega + 8\mu\,\omega + 9\mu^2, \tag{22}$$

$$C_{\rm p}(\omega, \mu) = -4\big(2^\nu - 2\big)(\omega + \mu)^2. \tag{23}$$

Effective structural coupling

Inspection of the invasion condition in Eq. (15) shows that structural heterogeneity and temporal reshuffling enter the criticality criterion through a single structural–temporal combination. Therefore, we define an effective structural coupling Q, which integrates excess membership, group-size heterogeneity, and plasticity:

$$Q = \left[\frac{\langle k(k-1) \rangle}{\langle k \rangle} + \frac{\langle k \rangle \, \omega}{\mu} \right] \frac{\langle n(n-1) \rangle}{\langle n \rangle}. \tag{24}$$

The first term inside the brackets captures the structural excess membership of nodes, while the second term accounts for the enhancement induced by temporal reshuffling over the contagion timescale μ^{-1}. The multiplicative factor involving $\langle n(n-1) \rangle$ encodes the excess group size, reflecting the number of potential reinforcement interactions within groups.

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